2023 Ph H2 Q1

Section: Our Dynamic Universe

Topic: Motion, Equations and Graphs

Summary of Question:

A van travels at 13.4 m s^{-1} and brakes with constant deceleration −2.85 m s^{-2} until coming to rest. Calculate: (a) the distance travelled during braking, (b) the time taken to stop, (c) complete the velocity–time graph.

(a) Distance travelled during braking

Equation: $s = (v^2 - u^2) / (2a)$

$s = (0^2 - 13.4^2) / (2 \times -2.85)$

$s = -179.56 / -5.70 = 31.5$ m

Answer: 31.5 m

(b) Time taken to stop

Equation: $v = u + at$

$0 = 13.4 + (-2.85)t$

$t = -13.4 / -2.85 = 4.70$ s

Answer: 4.7 s

(c) Velocity–time graph

- Straight line with negative gradient (−2.85 m s^{-2})

• From (0, 13.4) to (4.7, 0)

Answer: Straight line graph as described

Final Answers

(a) 31.5 m

(b) 4.7 s

(c) Straight line graph from (0,13.4) to (4.7,0)

Revision Tips

• Use SUVAT equations for constant acceleration problems.

• Remember: deceleration means negative acceleration.

• Velocity–time graph: gradient = acceleration; area = displacement.

• Both $s = (v^2 - u^2)/(2a)$ and $s = ut + \frac{1}{2}at^2$ can be used.